

Thermal Framework Corner Cases: Bugs or Undefined Behavior?

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Thermal zone names ?

- For x86 -> thermal zone type == CPU, NPU, GPU
- For DT based -> thermal zone type == name
- Finding out is a thermal is complicated:
  - /sys/class/thermal/thermal_zone??
  - Depending on the initialization order



- Proposal:
  - Introduce a thermal zone with name API
  - Add a link in pointing to the thermal_zoneXX
    - Eg. /sys/class/thermal/zones/cpu4 -> /sys/class/thermal/thermal_zone23



Thermal trip ordering consistency

- There is no check with the ordering vs types
  - Critical trip point can be before mitigation trip points leading to a reboot

- https://lore.kernel.org/all/20260710-critical_trip-v1-1-780d0b40bb96@oss.qualcomm.com/
  - code not optimized, double walk
  - message not useful
  - What to do ?





Step wise governor

- High number of states
  - Not friendly with RPM fan
  - Too long to reach the max state (too many loops)
- +1 / -1 only, may be too slow

- Introduce LUT in step wise ? (prevents driver code duplication)





Polling delay

- Copied / pasted values from another platform
- Depending on the thermal situation, temperature transitions can be faster at the same trip point

- Adaptative polling delay (Shannon theorem)?
  - Measure the up / down interval and the speed
  - Increase / decrease the polling delay ?



Suspend / Resume

- If suspend during mitigation, at resume time if the temperature dropped below the trip it is not detected and no notification is sent

- That was observed but the opposite may happen if the temperature is above the trip point during suspend

